ELE 3125 - Signals & Systems

Lecture 9 - Part 1

Fourier series Representation of Discrete-Time Periodic Signals

Linear Combination of Harmonically Related Complex Exponentials

$x[n] = x[n+N]$

$\omega_0 = \frac{2\pi}{N}$

$\phi_k[n] = e^{jk\omega_0 n} = e^{jk(2\pi/N)n}$, $k = 0, \pm 1, \pm 2, \ldots$,

Only N distinct signals in the set given by the above equation

$$e^{j(k+rN)\omega_0 n} = e^{jk\omega_0 n} e^{jrN\omega_0 n} = e^{jk\omega_0 n} e^{jr2\pi n} = e^{jk\omega_0 n} \underbrace{e^{jr2\pi n}}_{=1} = e^{jk\omega_0 n}$$

$$\Rightarrow \boxed{\phi_k[n] = \phi_{k+rN}[n]}$$

Representation of periodic sequences in terms of linear combinations of the sequences $\phi_k[n]$

$$x[n] = \sum_k a_k \underbrace{\phi_k[n]}_{\downarrow} = \sum_k a_k e^{jk\omega_0 n} = \sum_k a_k e^{jk(2\pi/N)n}$$

over a range of N successive values of k $\Rightarrow k = \langle N \rangle$

$$x[n] = \sum_{k=\langle N \rangle} a_k \phi_k[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

Determination of the Fourier Series Representation of a Periodic Signal

DTFS pair :

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} \qquad \text{synthesis equation}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} \qquad \text{analysis equation}$$

CTFS pair :

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t} \qquad \text{synthesis equation}$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt \qquad \text{analysis equation}$$

Example: Consider the signal

$x[n] = 1 + \sin\left(\frac{2\pi}{N}\right)n + 3\cos\left(\frac{2\pi}{N}\right)n + \cos\left(\frac{4\pi}{N}n + \frac{\pi}{2}\right)$, Recall: $\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$, $\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$

$= 1 + \left[\frac{e^{j(2\pi/N)n} - e^{-j(2\pi/N)n}}{2j}\right] + 3\left[\frac{e^{j(2\pi/N)n} + e^{-j(2\pi/N)n}}{2}\right] + \frac{e^{j\left[\frac{4\pi}{N}n + \frac{\pi}{2}\right]} + e^{-j\left[\frac{4\pi}{N}n + \frac{\pi}{2}\right]}}{2}$

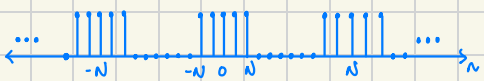
$= 1 + \frac{1}{2j}e^{j(2\pi/N)n} - \frac{1}{2j}e^{-j(2\pi/N)n} + \frac{3}{2}e^{j(2\pi/N)n} + \frac{3}{2}e^{-j(2\pi/N)n} + \left(\frac{e^{j(4\pi/N)n}e^{j(\pi/2)} + e^{-j(4\pi/N)n}e^{-j(\pi/2)}}{2}\right)$

$= 1 + \frac{1}{2j}e^{j(2\pi/N)n} + \frac{3}{2}e^{j(2\pi/N)n} - \frac{1}{2j}e^{-j(2\pi/N)n} + \frac{3}{2}e^{-j(2\pi/N)n} + \frac{e^{j(\pi/2)}}{2}e^{j2(2\pi/N)} + \frac{e^{-j(\pi/2)}}{2}e^{-j2(2\pi/N)}$

$= \underbrace{1}_{} + \underbrace{\left(\frac{3}{2} + \frac{1}{2j}\right)}e^{j(2\pi/N)n} + \underbrace{\left(\frac{3}{2} - \frac{1}{2j}\right)}e^{-j(2\pi/N)n} + \underbrace{\frac{1}{2}j}e^{j2(2\pi/N)} - \underbrace{\frac{1}{2}j}e^{-j(2)(2\pi/N)}$

Fourier series coefficients: $\boxed{a_0 = 1,\ a_1 = \frac{3}{2} + \frac{1}{2j},\ a_{-1} = \frac{3}{2} - \frac{1}{2j},\ a_2 = \frac{1}{2}j,\ a_{-2} = -\frac{1}{2}j}$

Example: Consider the square wave shown in the figure below (Note: it is very similar to a continuous-time periodic square wave.



$a_k = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]e^{-jk(2\pi/N)n} = \frac{1}{N}\sum_{n=-N_1}^{N_1} x[n]e^{-jk(2\pi/N)n}$, let $m = n + N_1 \Rightarrow n = m - N_1$, when $n = -N_1 \Rightarrow m - N_1 = -N_1 \Rightarrow m = N_1 - N_1 = 0$

$\Rightarrow$ when $n = N_1 \Rightarrow m - N_1 = N_1 \Rightarrow \boxed{m = 2N_1}$ $\Rightarrow \boxed{m = 0}$

$= \frac{1}{N}\sum_{m=0}^{2N_1} \underbrace{x[n]}_{=1} e^{-jk(2\pi/N)(m-N_1)} = \frac{1}{N}\sum_{m=0}^{2N_1} e^{-jk(2\pi/N)m}e^{jk(2\pi/N)N_1} = \frac{1}{N}e^{jk(2\pi/N)N_1}\sum_{m=0}^{2N_1} e^{-jk(2\pi/N)m}$

$= \frac{1}{N}e^{jk(2\pi/N)N_1}\sum_{m=0}^{2N_1}\left[e^{-jk(2\pi/N)}\right]^m$, In general: $\sum_{k=N_1}^{N_2} a^k = \frac{a^{N_1} - a^{N_2+1}}{1-a}$

$= \frac{1}{N}e^{jk(2\pi/N)N_1}\left[\frac{\left(e^{-jk(2\pi/N)}\right)^0 - \left(e^{-jk(2\pi/N)}\right)^{(2N_1+1)}}{1 - \left(e^{-jk(2\pi/N)}\right)}\right] = \frac{1}{N}e^{jk(2\pi/N)N_1}\left[\frac{1 - e^{-jk2\pi(2N_1+1)/N}}{1 - e^{-jk(2\pi/N)}}\right]$

$= \frac{1}{N}e^{jk(2\pi/N)N_1}\left[\frac{1 - e^{-jk(2\pi)(2)(N_1+\frac{1}{2})/N}}{e^{-jk(2\pi/N)(\frac{1}{2})}\left(e^{jk(2\pi/N)(\frac{1}{2})} - e^{-jk(2\pi/N)(\frac{1}{2})}\right)}\right]$

$= \frac{1}{N}e^{jk(2\pi/N)N_1}\left[\frac{e^{-jk2\pi(N_1+\frac{1}{2})/N}\left(e^{jk2\pi(N_1+\frac{1}{2})/N} - e^{-jk2\pi(N_1+\frac{1}{2})/N}\right)}{e^{-jk(2\pi/N)(\frac{1}{2})}\left(e^{jk(2\pi/N)(\frac{1}{2})} - e^{-jk(2\pi/N)(\frac{1}{2})}\right)}\right]$

$= \frac{1}{2}e^{jk(2\pi/N)N_1}\frac{\left(e^{-jk2\pi(N_1+\frac{1}{2})/N}\right)}{\left(e^{-jk(2\pi/N)(\frac{1}{2})}\right)}\left[\frac{\left(e^{jk2\pi(N_1+\frac{1}{2})/N} - e^{-jk2\pi(N_1+\frac{1}{2})/N}\right)/2j}{\left(e^{jk(2\pi/N)(\frac{1}{2})} - e^{-jk(2\pi/N)(\frac{1}{2})}\right)/2j}\right]$

$= \frac{1}{2}\frac{\overbrace{e^{jk(2\pi/N)N_1}e^{-jk(2\pi/N)N_1}}^{=1}\cancel{e^{-jk(\frac{2\pi}{N})(\frac{1}{2})}}}{\cancel{e^{-jk(\frac{2\pi}{N})(\frac{1}{2})}}}\left[\frac{\left(e^{jk2\pi(N_1+\frac{1}{2})/N} - e^{-jk2\pi(N_1+\frac{1}{2})/N}\right)/2j}{\left(e^{jk(2\pi/N)(\frac{1}{2})} - e^{-jk(2\pi/N)(\frac{1}{2})}\right)/2j}\right] = \frac{1}{2}\frac{\sin\left(2\pi k(N_1+\frac{1}{2})/N\right)}{\sin(\pi k/N)}$

$\Rightarrow \boxed{a_k = \frac{1}{2}\frac{\sin\left(2\pi k(N_1+\frac{1}{2})/N\right)}{\sin(\pi k/N)}}$

A couple more properties

Multiplication

$$x[n]y[n] \overset{FS}{\longleftrightarrow} \sum_{l=\langle N \rangle} a_l b_{k-l}$$

First Difference

$$x[n]-x[n-1] \overset{FS}{\longleftrightarrow} \left(1-e^{-jk(2\pi/N)}\right) a_k$$

Parseval's Relation

$$\frac{1}{T}\sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$